

MINUGA KARTHIK KUMAR

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AI/GenAI systems engineer who builds and ships working LLM applications end to end. Architected a tool-using coding agent with a think → act → observe loop and a Tree-sitter code knowledge graph that cuts LLM token consumption by up to 60% while preserving semantic accuracy, and an AI-assisted security platform that uses the Claude API to turn raw scanner output into developer-ready remediation. Works across Python and TypeScript — FastAPI/Node backends, Dockerized deployments, local LLM pipelines with Ollama, semantic retrieval over vector stores, and pytest coverage — with a track record of self-initiated projects taken from problem selection and spec writing through debugging and deployment.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, C

Generative AI: LLM APIs (Claude, Copilot vscode.Im, OpenAI-compatible), Prompt Engineering, AI Agents, Tool Calling, Agent Loops, Context Engineering, Agent Memory Architectures, Semantic Retrieval, Local LLMs (Ollama, Llama-family), Model Fine-Tuning

AI / ML: Machine Learning, Deep Learning, Computer Vision, YOLO / Ultralytics, CNNs, DistilBERT, sentence-transformers, Data Preprocessing & Model Evaluation

AI Application Development: FastAPI, Flask, Node.js, REST APIs, Celery, Redis, Streamlit, VS Code Extension API, Tree-sitter (WASM), FFmpeg

Data & Storage: PostgreSQL, ChromaDB, MongoDB, MySQL, SQLite

Developer Tools & Deployment: Git, GitHub, Docker, Docker Compose, Linux, pytest, Nginx, Grafana

Security Engineering: Vulnerability Assessment / VAPT, OWASP Top 10, Application Security, OWASP ZAP, Nmap, Nikto, Burp Suite

PROJECTS

NexCode — AI-Powered Coding Agent for VS Code | *TypeScript, Node.js, VS Code Extension API, Tree-sitter WASM, LLM APIs*

- Architected a graph-based code representation engine using Tree-sitter WASM that compresses an entire codebase into a structured knowledge graph, cutting LLM token consumption by up to 60% while preserving semantic accuracy — context engineering rather than raw file dumping.
- Engineered dependency edge mapping across imports, function calls, class hierarchies, and cross-file relationships, so the model reasons reliably over large codebases inside a compressed context window.
- Designed a think → act → observe agent loop with five extensible tools (code generation, execution, debugging, explanation, file read/write), orchestrating structured and auditable interaction between the LLM and the local development environment.
- Self-initiated and owned end to end: problem selection, architecture and specification, extension packaging, and debugging of the agent loop against real editor workflows.

VAPTaaS — AI-Assisted Security Assessment Platform | *FastAPI, Celery, Redis, PostgreSQL, Docker Compose, Claude API, React, pytest*

- Architected an asynchronous, multi-stage pipeline orchestrating reconnaissance (Nmap), web vulnerability scanning (OWASP ZAP), and misconfiguration checks (Nikto) through Celery task queues backed by Redis and PostgreSQL.
- Integrated the Claude API as a reasoning layer that transforms raw, unstructured tool output into contextual, developer-readable remediation guidance, converting scan noise into actionable engineering work.
- Containerized the full stack with Docker Compose for repeatable deployments and implemented a pytest suite covering scan orchestration and API endpoints; debugged async failure modes including scan timeouts, database driver issues, and report-generation crashes.

Agentic Memory Architecture — Local-LLM Agent Runtime (Experimental) | *Python, Ollama, Llama-family models, ChromaDB, sentence-transformers, OpenClaw runtime*

- Designed a hybrid memory architecture combining Markdown stores with ChromaDB, using sentence-transformer embeddings for semantic retrieval — a retrieval-oriented context assembly layer feeding a local LLM agent.
- Modeled distinct episodic, semantic, reflection, and operational memory types to support agent reasoning, planning, and tool use, with CLI-based memory management for inspection and curation.
- Prototype research project, built to test how retrieval strategy and memory structure change agent behavior over long-running sessions.

Talking Avatar — Local LLM Multimodal Pipeline | *Python, Ollama, gTTS, pydub, FFmpeg, Wav2Lip*

- Built an offline pipeline chaining local LLM response generation → speech synthesis → lip-synced video, running entirely on Ollama-hosted models with no external API dependency.
- Debugged and stitched together multimodal tooling (gTTS, pydub, FFmpeg, Wav2Lip) into a single reproducible generation workflow.

Crack Detection & Severity Analysis System | *Python, YOLO (v5/v8), CNNs, Streamlit, Computer Vision*

- Engineered an end-to-end image-processing and reporting pipeline for automated crack identification and severity classification in civil infrastructure, deployed through Streamlit for real-time inspection workflows.
- Experimented with and evaluated YOLOv5 and YOLOv8 models to select the detection backbone for the deployed pipeline.

Security Tooling — Distributed Honeypot Platform & Security Header Analyzer | Python, FastAPI, PostgreSQL, Docker, Grafana, Streamlit, ReportLab

- Architected a multi-service deception platform with independently containerized web, FTP, and API honeypots, normalizing raw attacker activity into structured alerts and attacker profiles surfaced through real-time dashboards.
- Built a multi-threaded HTTP security header auditor scanning up to 50 domains concurrently, with scan-history comparison and automated PDF/JSON reporting.

Additional: Flutter + Flask translation application using Argos Translate for offline neural machine translation, with Whisper speech-to-text integration prototyped.

EXPERIENCE

AI / Machine Learning Intern — Viswa Robotic Cars March 2026 – Present

- Contributing to AI and machine learning projects spanning intelligent system development and model implementation, iterating with the team on Python-based ML workflows.

Machine Learning Intern — Viswa Robotic Cars June 2025 – August 2025

- Worked on data preprocessing, model development, and evaluation, collaborating with mentors on industry-oriented AI development workflows.

Google Android Developer Intern — EduSkills × AICTE March 2026 – June 2026

- Developed Android applications with Android Studio and Git, following structured software development and review workflows.

Cybersecurity Intern — Cisco Networking Academy June 2025 – August 2025

- Completed practical training in network security, cyber defense, and threat analysis fundamentals.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) 2023 – 2027

Madanapalle Institute of Technology & Science — CGPA: 8.03 / 10.0

CERTIFICATIONS & ACHIEVEMENTS

- Cyber Security Professional Program (240 Hrs) — Ethnotech Academy Pvt. Ltd.; ethical hacking, web/API security, SOC operations & SIEM, cloud security fundamentals.
- Cybersecurity Essentials & Network Basics — Cisco Networking Academy | Fundamentals of Cybersecurity — Coursera | Digital 101 Gold Certificate — NASSCOM.
- Winner, Department-Level Project Expo | HackerRank 3-Star Badge in C (Data Structures & Algorithms).
- Event Coordinator, AIMEX 2K25 — ran end-to-end logistics and participant coordination for a departmental technical fest; Coordinator, IQNITE quiz platform challenge.